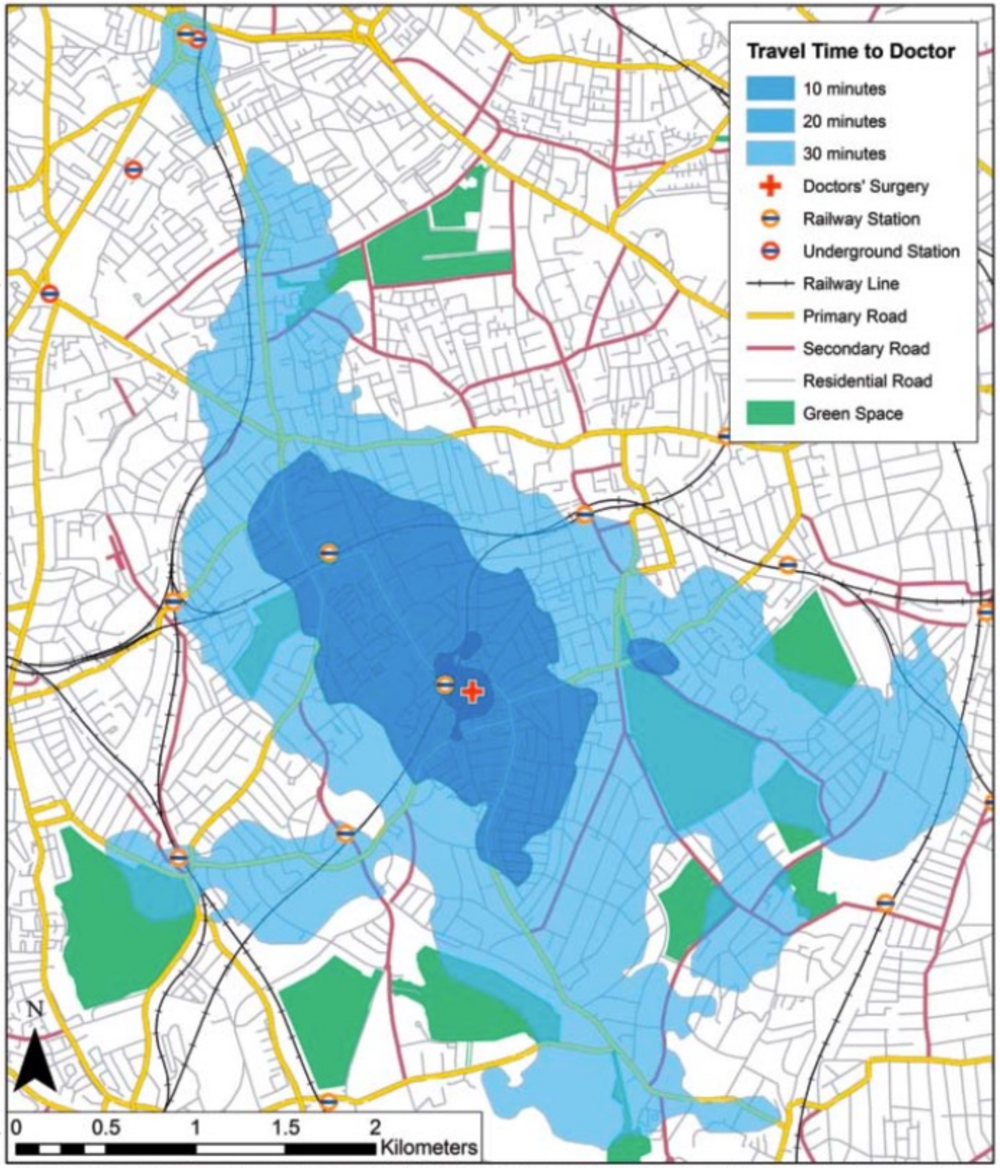
Geographic Representation

GEOG 1026

Department of Geography

National Taiwan University

Longley et al.



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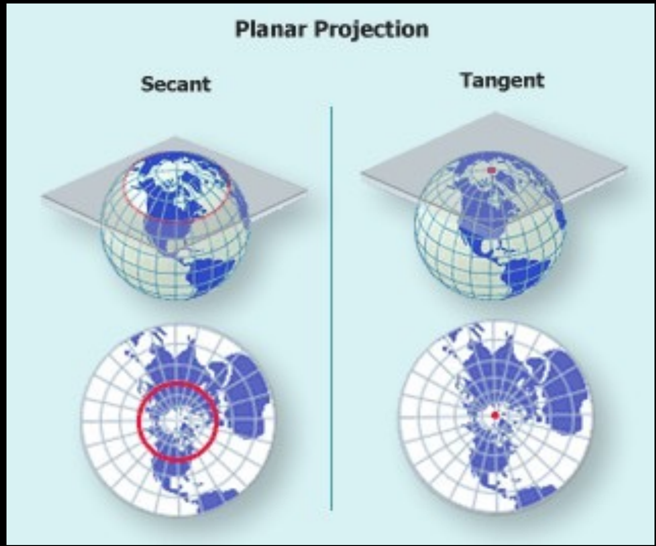
Map author: Daniel Lewis, UCL Department of Geography, May 2009

Last Week

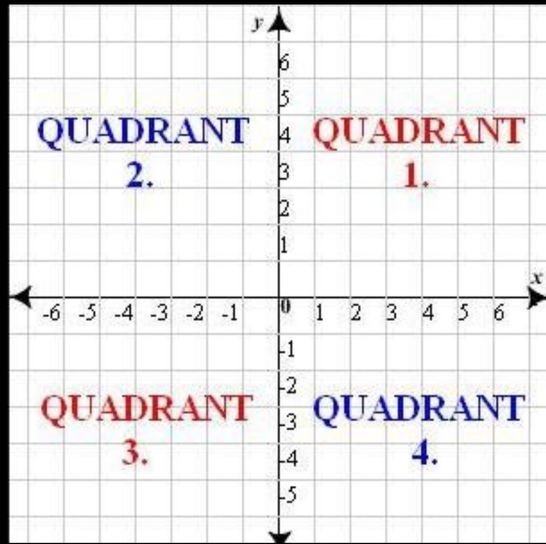
- Geodesy (測地學、大地測量學)
- Figure of Earth (地球形狀)
- Coordinate system (坐標系統)
- Map projection (地圖投影)

Key Concept

Geographic
Coordinate System

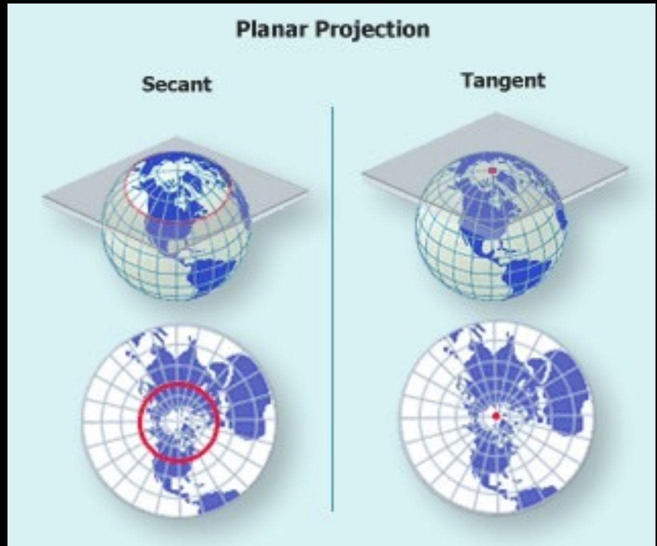


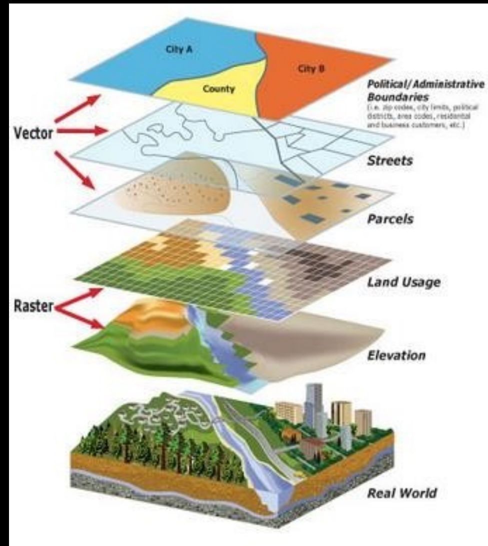
Map
Projection



Plane
Coordinate

Key Concept





Geographic Coordinate System

Map Projection

Geographic Data

Spherical coordinates

Plane coordinates

This Week

- The nature of geographic data
  - Spatial autocorrelation (空間自相關), distance decay (距離衰減), self-similarity (自我相似性), and scale (尺度)
  - Spatial sampling (空間取樣)

- Geographic Representation
  - Representation: Discrete objects (離散物件) and continuous fields (連續場)
  - Digital representation: raster data (網格資料) and vector data (向量資料)
  - Generalization (概括化)

Support Infrastructure for Decision-Making

Decision-making support infrastructure	Ease of sharing with everyone	GIS example
Wisdom ↑	*Impossible*	Policies developed and accepted by stakeholders
Knowledge ↑	*Difficult, especially tacit knowledge*	Personal knowledge about places and issues
Evidence ↑	*Often not easy*	Results of GIS analysis of many data sets or scenarios
Information ↑	*Easy*	Contents of a database assembled from raw facts
Data	*Easy*	Raw geographic facts

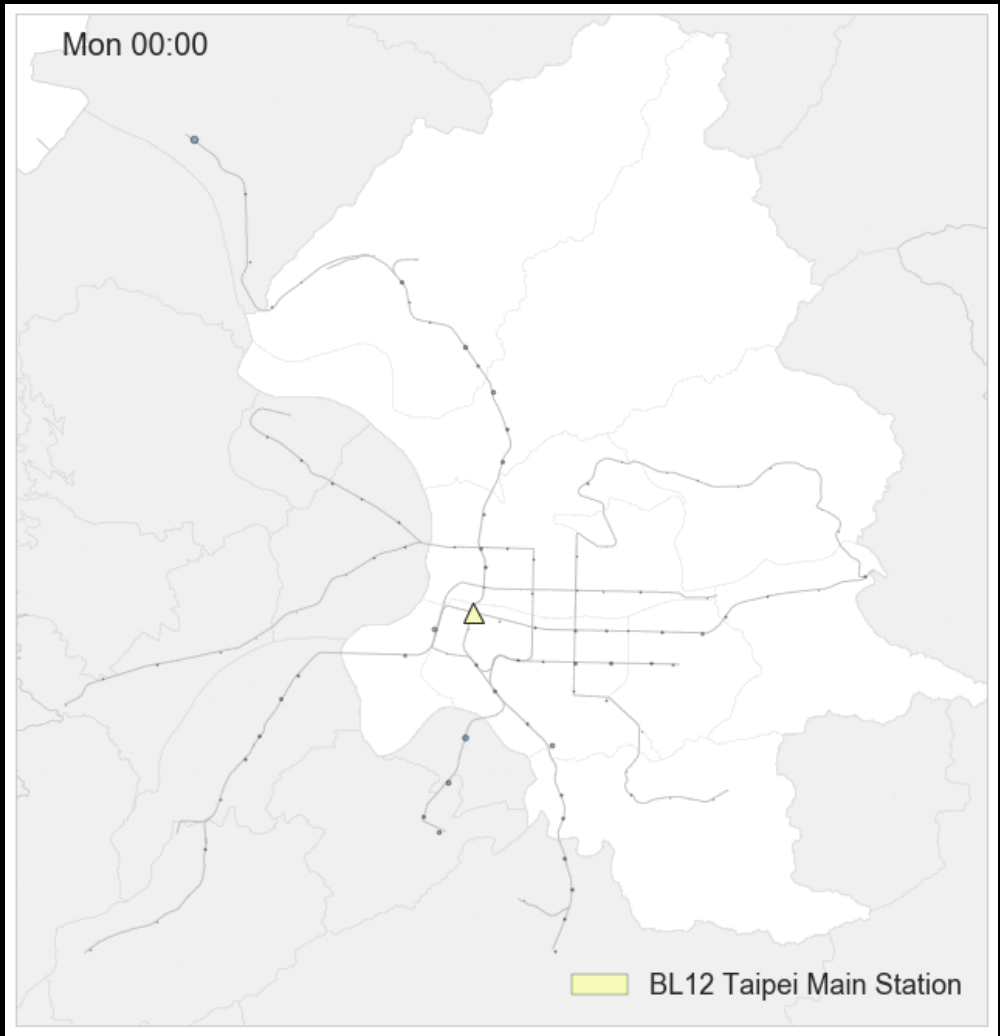
Longley et al. (2014)

- The central motivation for geographic information (GI) science as the development of representations, not only of how the world **looks**, but also how it **works**.

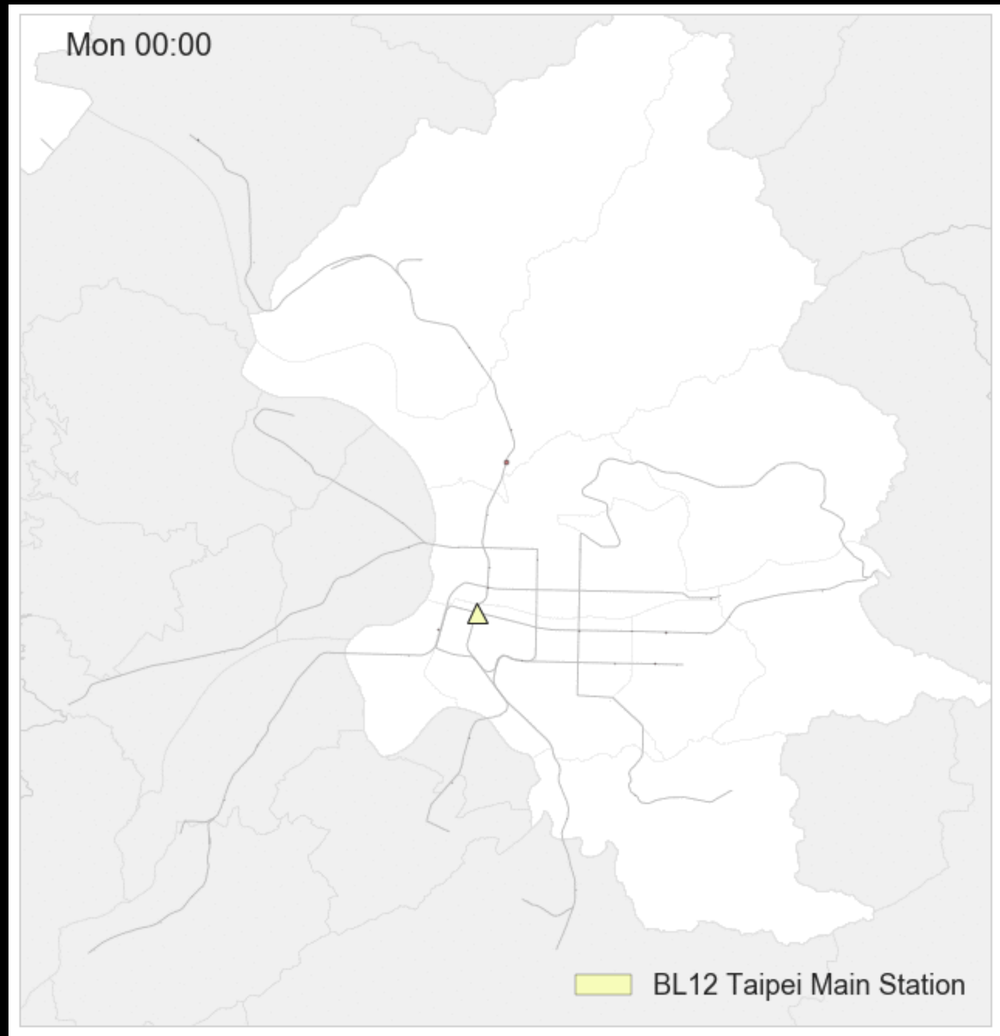
The Fundamental Problem

- How to describe your life?

人都從台北車站搭去哪?



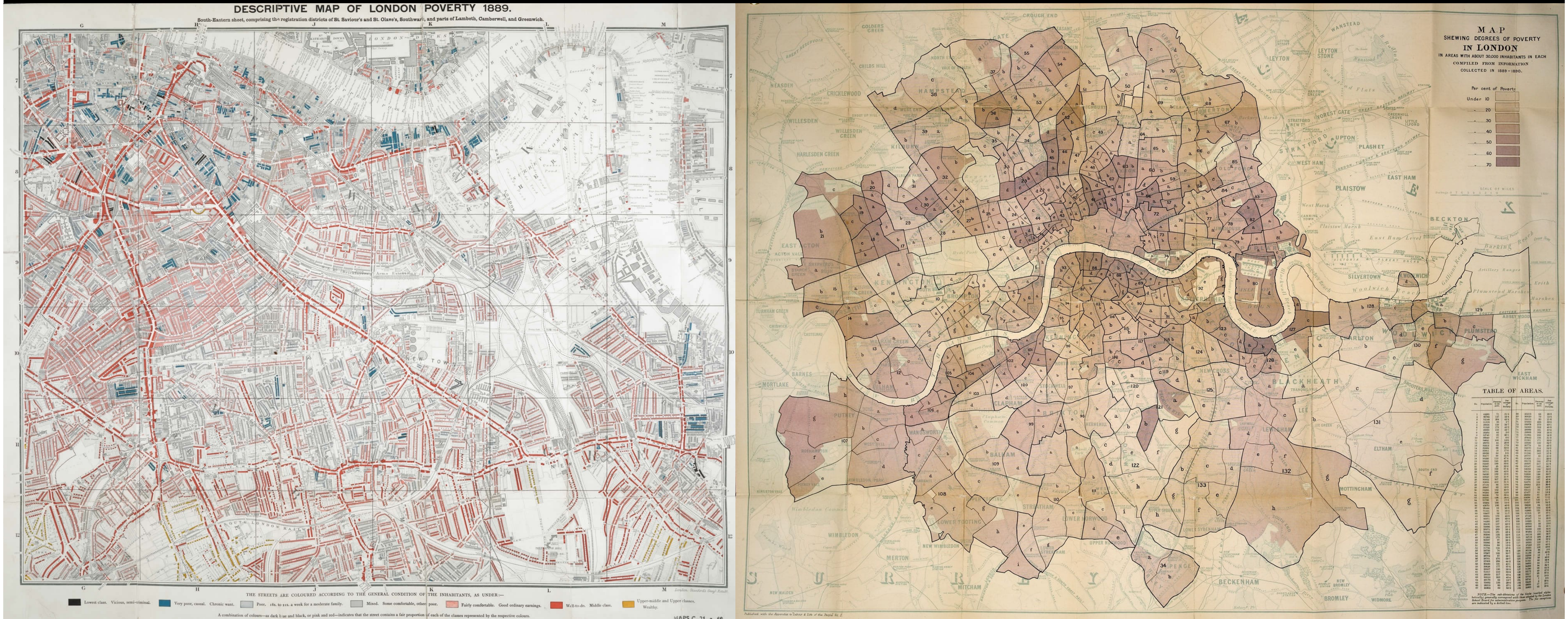
人都從哪搭去台北車站?



https://medium.com/geopainter

W1

Is urban poverty caused by individuals?



Descriptive Map of London Poverty, 1889, Charles Booth

The Fundamental Problem

- GI scientists distinguish between controlled and uncontrolled variation over time.
- Our behavior in geographic space often reflects past pattern of behavior.
- Explanation in time need only look to the past, but explanation in space must look in all directions simultaneously.
- Temporal autocorrelation will be strongest between events that happen at about the same time as one another, so we expect spatial autocorrelation (空間自相關) to be stronger for occurrences that are located close to one another.

W1

Laws of Geography

- Tobler’s first law of geography:
  - Everything is related to everything else, but near things are more related than distant.

- Tobler’s second law of geography:
  - The phenomenon external to a geographic area of interest affects what goes inside.

- Arbia law of geography:
  - Everything is related to everything else, but things observed at a coarse spatial solution are more related than things observed at a finer resolution.

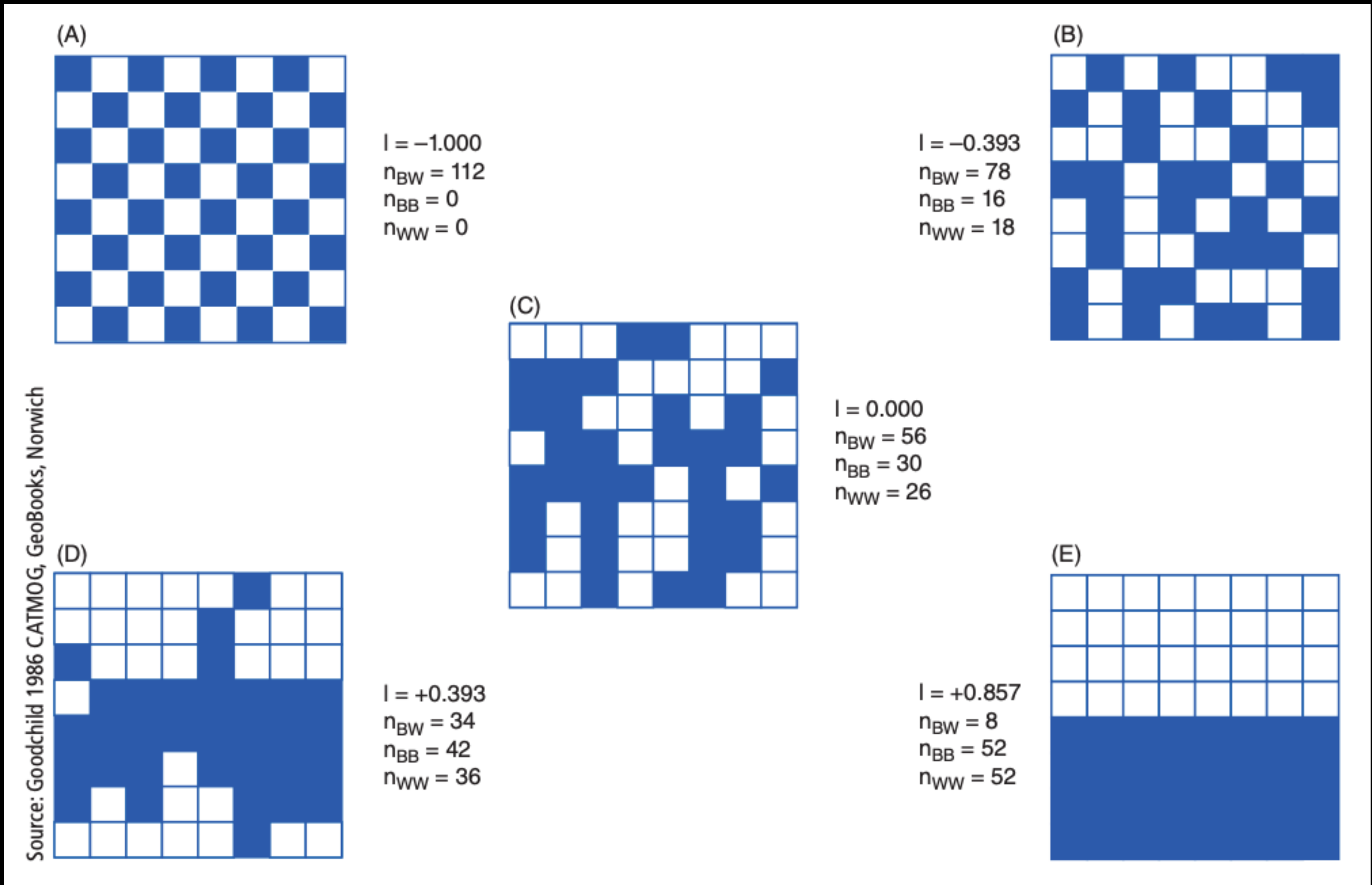
The Nature of Geographic Data: Spatial Autocorrelation

- Tobler's first law of geography:
  - Everything is related to everything else, but near things are more related than distant.

- Spatial data exhibit an increasing range of values, hence increased heterogeneity with increased distance.

- Spatial autocorrelation measures attempt to deal simultaneously with similarities in the location of spatial objects and their attributes.

The Nature of Geographic Data: Spatial Autocorrelation



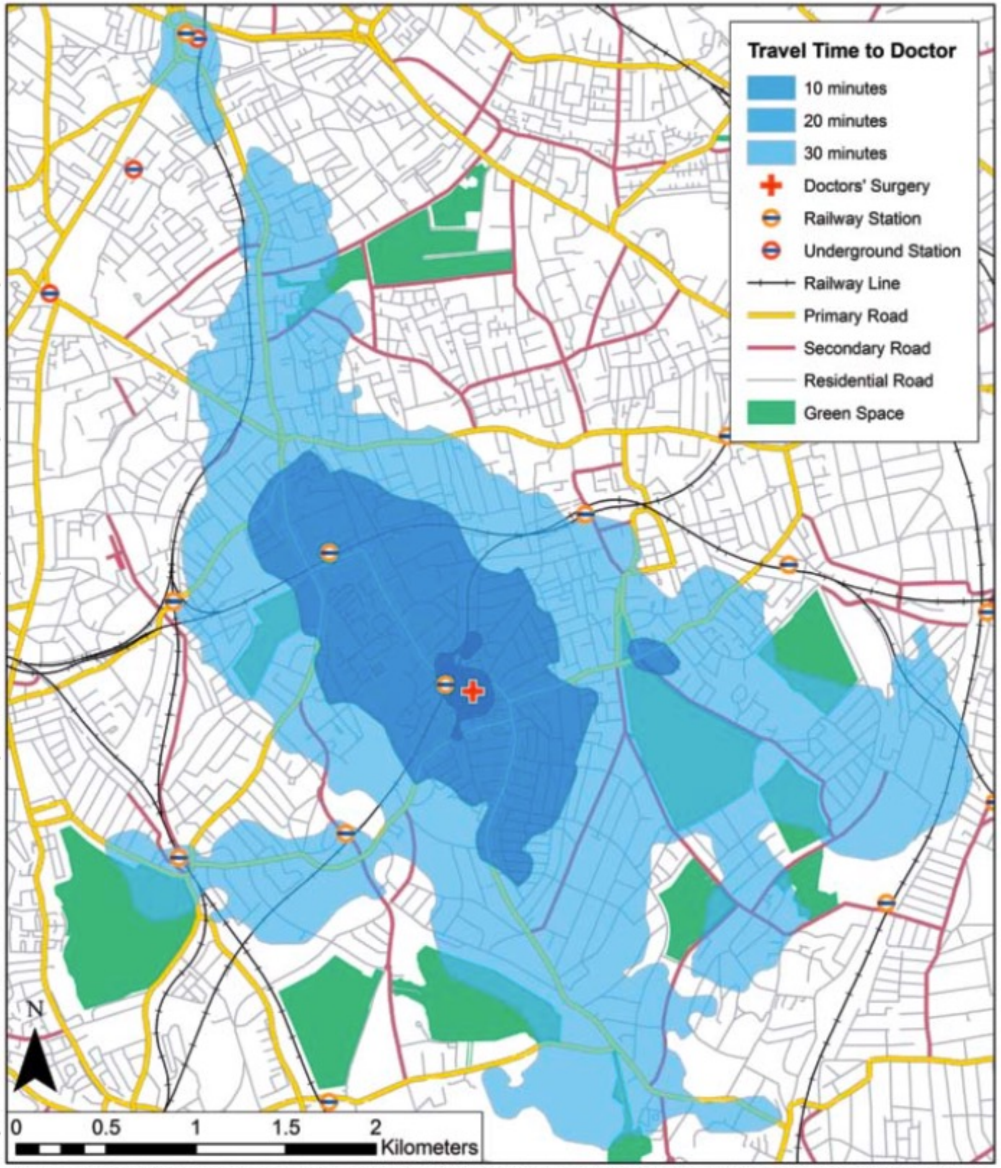
(A) Extreme negative spatial autocorrelation

(B) A dispersed arrangement

(C) Spatial independence

(D) Spatial clustering

(E) Extreme positive spatial autocorrelation



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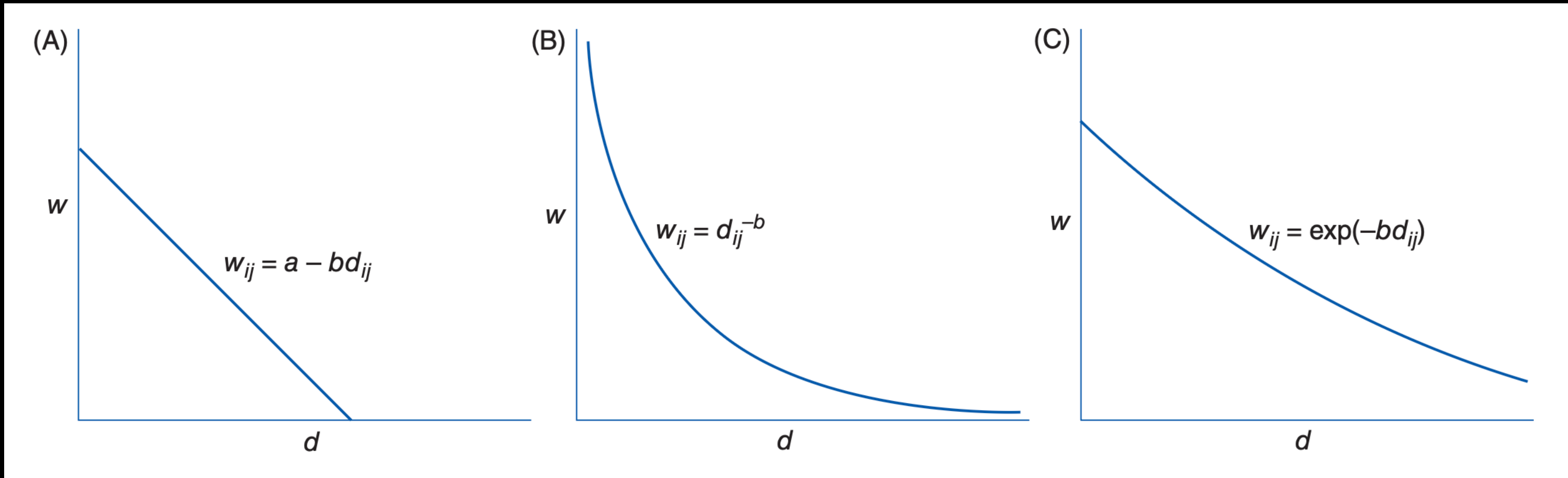
Longley et al. (2014)

The Nature of Geographic Data: Distance Decay

Linear decay function | Negative power distance decay function | Negative exponential statistical fit

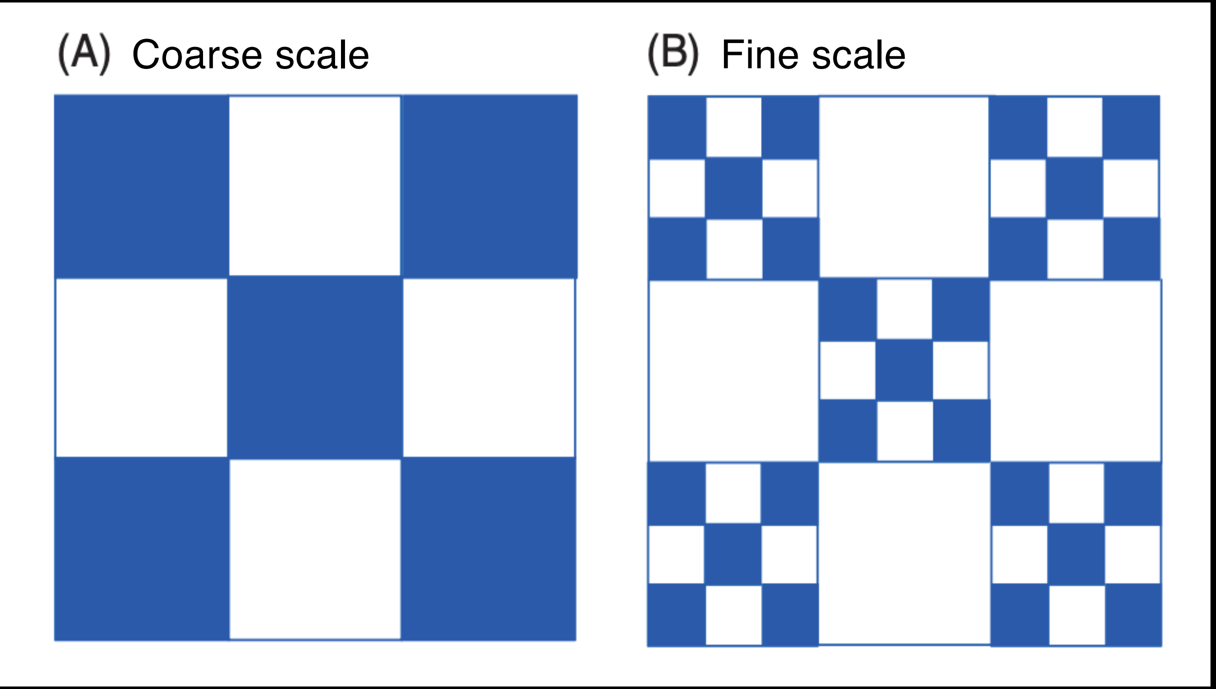


- We need to specify likely attenuation effect (衰減效應) of distance.
- We need to make an informed judgement about appropriate interpolation (空間內插) function and how to weight (加權) adjacent observation.

The Nature of Geographic Data: Autocorrelation

- However,...
- Some geographic phenomena vary smoothly across space, whereas others can exhibit extreme irregularity, in violation of Tobler's Law.
- It is highly likely that a representation of the real world that is suitable for predicting future change will need to incorporate information on how two or more factors covary (共變).
- Spatial autocorrelation helps us to build representations but frustrates our efforts to predict.

The Nature of Geographic Data: Self-Similarity



- The pattern of spatial autocorrelation at the coarser scale is replicated at the finer scale, and the overall pattern is said to exhibit the property of self-similarity (自我相似性), such as individual rocks may resemble the forms of larger structures.

The Nature of Geographic Data: Self-Similarity

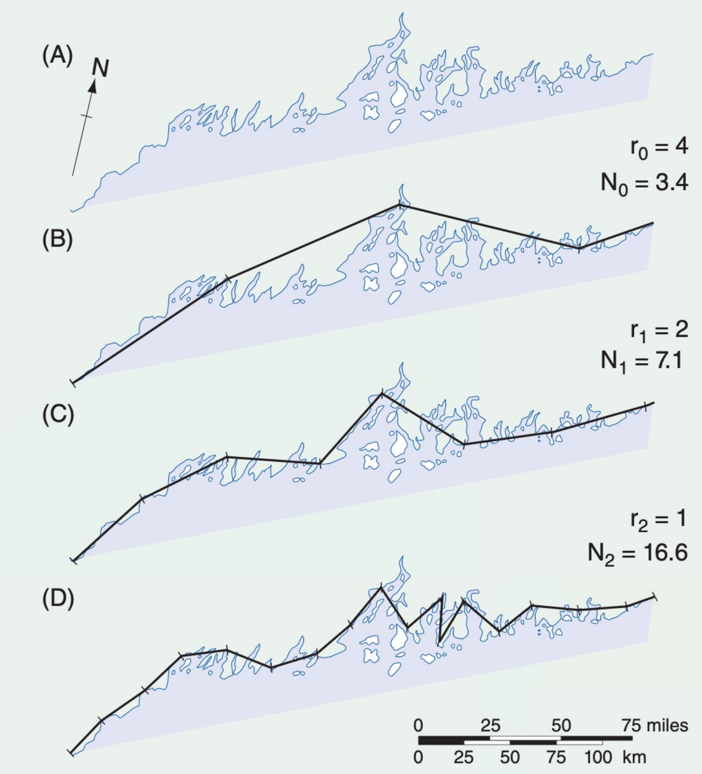
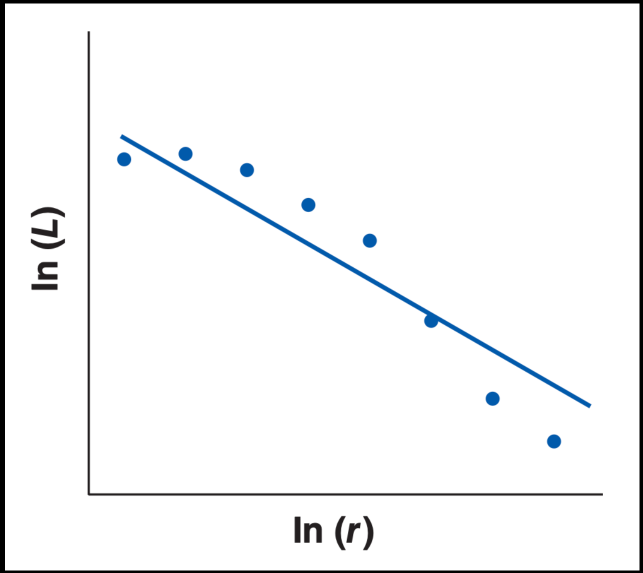


Figure 2.14 The coastline of Maine, at three levels of recursion: (A) the base curve of the coastline, (B) approximation using 100-km steps, (C) 50-km step approximation, and (D) 25-km step approximation.



The relationship between recorded length (L) and step length (r)

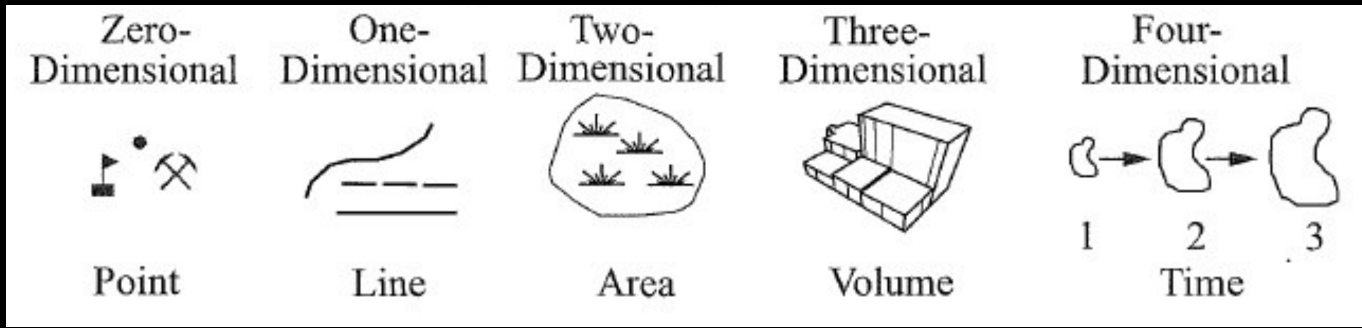
- In a self-similar object, each part has the same structure as the whole.
- Much spatial variation has jagged irregularity, which is often observed across a range of scales.
- A more general geometry of the irregular, termed fractal geometry (碎形), has come to provide a more appropriate and general means of summarizing the structure and character of spatial objects.

The Nature of Geographic Data: Scale

- The issue of sampling interval is of direct importance in the measurement of spatial autocorrelation.

- Scale is often integral to the trade-off between the level of spatial resolution and the degree of attribute detail that can be stored in a given application.

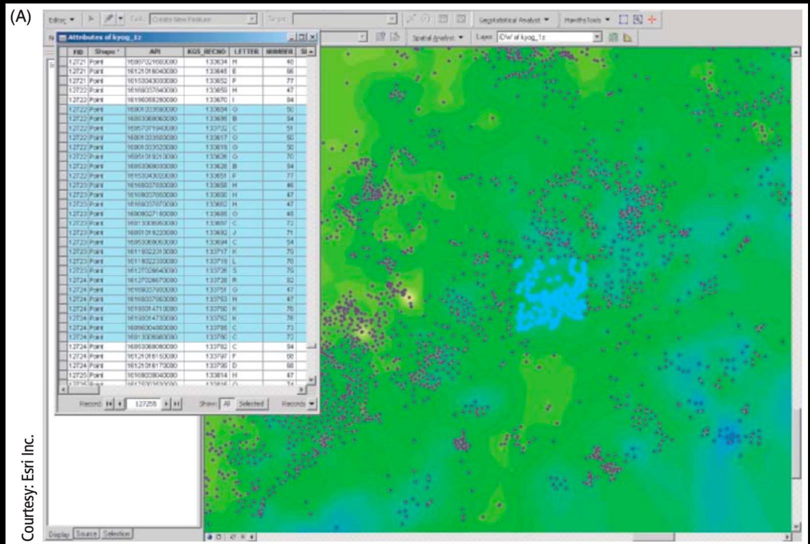
Types of Spatial Objects



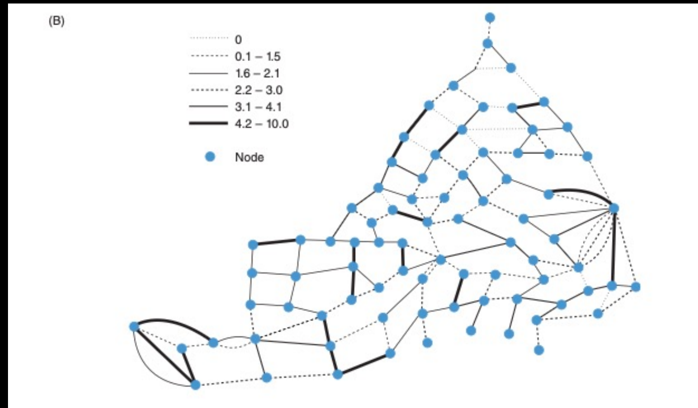
- Geographic objects are classified according to their topological dimension, which provide a measure of the way they fill space.
- A point has neither length nor breadth nor depth, and hence it is said to be of dimension 0.
- Lines have length, but not breadth or depth and hence are of dimension 1.
- Area objects have the two dimensions of length and breadth, but not depth.
- Volume objects have length, breadth, and depth and hence are of dimension 3.
- Time is often considered to be the fourth dimension of spatial objects.

Types of Spatial Objects

0D

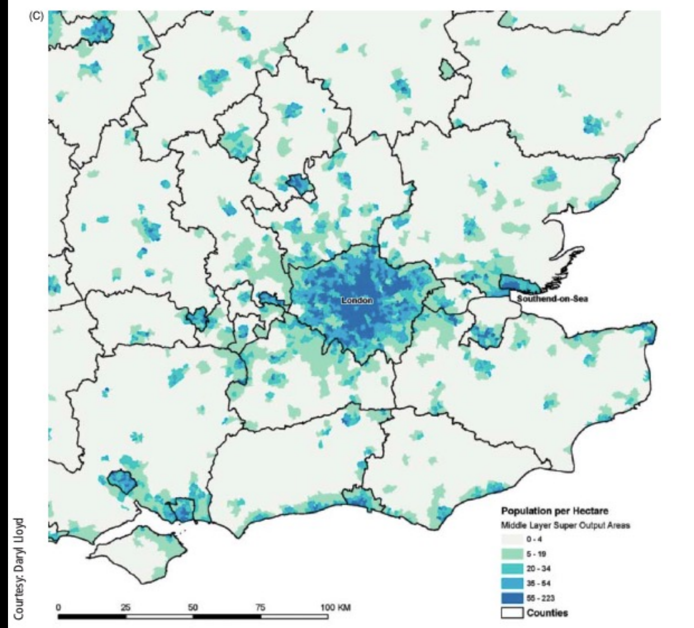


1D



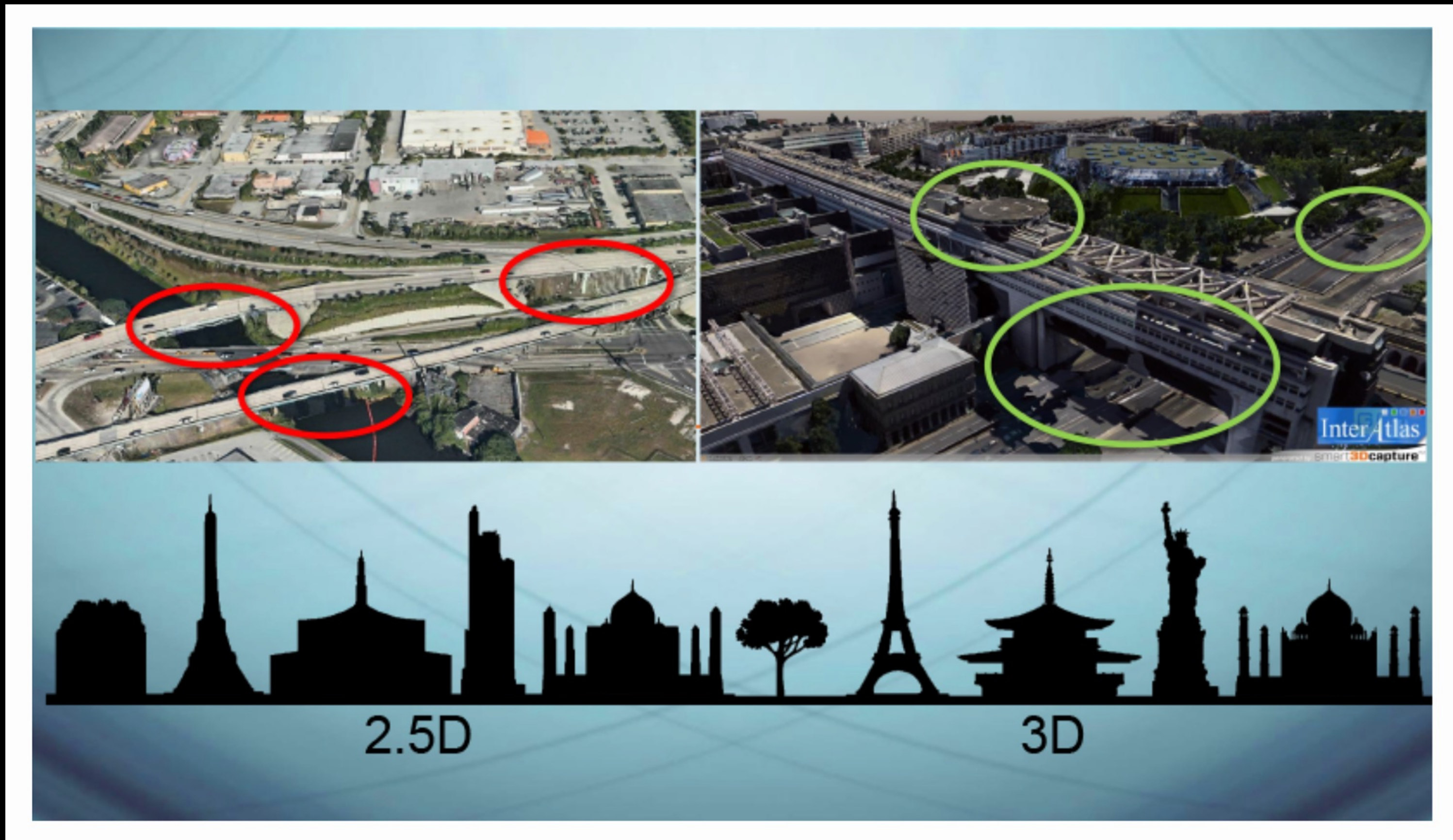
3D

2D

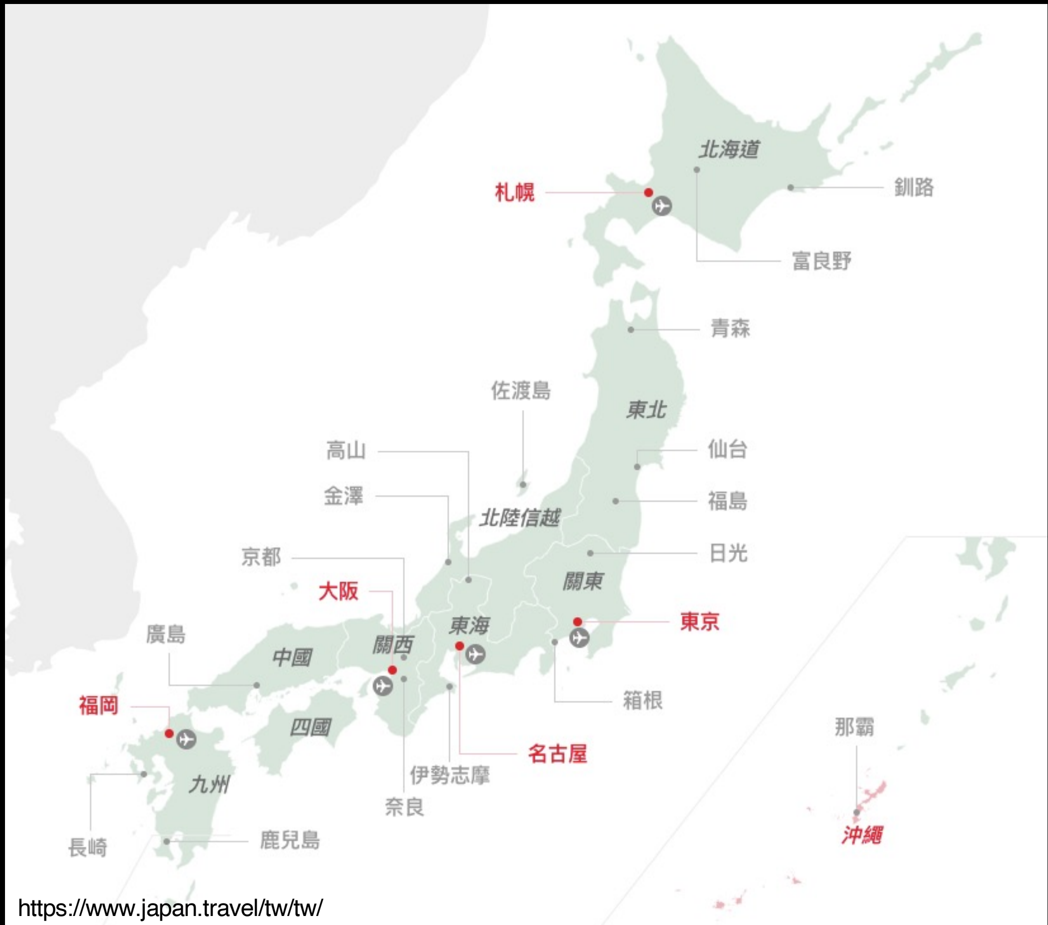


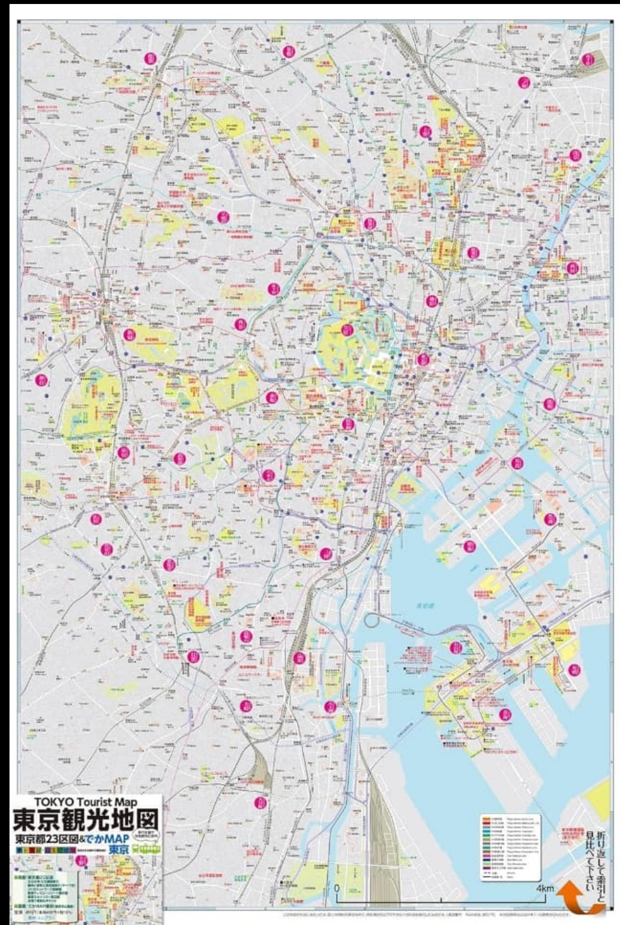
Types of Spatial Objects

- GI users sometimes talk about 2.5D rather than 3D.



http://www.linkfast.com.tw/index.php?action=product_detail&cid=20&ccid=54&id=50





- The classification of spatial phenomena into object types is dependent fundamentally on scale.

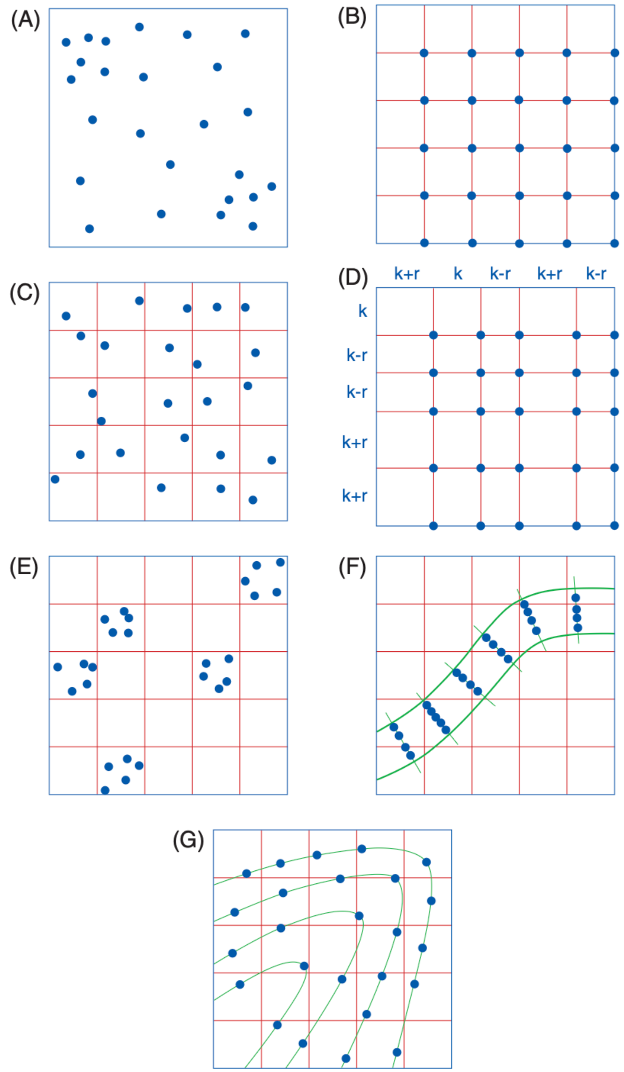
Spatial Sampling (空間抽樣)

- The quest to generalize about the myriad complexity of the real world requires us to abstract, or sample, events and occurrences from the universe of eligible elements of interest, which is known as the sample frame (樣本框架).
- Geographic data area only as good as the sampling scheme used to create them.
- A statistical surface of geographic phenomena is estimated through spatial sampling.
- Random sampling (隨機抽樣) enables us to use the distribution of values in our sample to tell us something about the like distribution of values in the parent population (母體).
- Randomly draw elements are disproportionately concentrated among some parts of the population at the expense of others, particularly when the size of our sample is small relative to the population.

Spatial Sampling (空間抽樣)

- Systematic spatial sampling aims to circumvent the problem and ensure greater evenness of coverage across the sample frame.
- Stratified sampling (分層抽樣) designs accommodate the unequal abundance of different phenomena on the Earth's surface.
- The advantage offers over simple random sampling may be two-edged, if the sampling interval and the spatial structure coincide.
- A number of hybrid sample designs have been devised to get around the vulnerability of spatially systematic sample designs to periodicity and the danger that simple random sampling may generate freak samples.

Figure 2.4 Spatial sample designs: (A) simple random sampling, (B) stratified sampling, (C) stratified random sampling, (D) stratified sampling with random variation in grid spacing, (E) clustered sampling, (F) transect sampling, and (G) contour sampling.



Big and Open Geographic Data

- More and more geographic data are collected by other parties, which may variously be nonexpert, self-selecting, or self-interested.
- Volunteered geographic information (VGI) is the collective harnessing of tools to create, assemble, and disseminate geographic data provided voluntarily by individuals.
- The content, coverage, and collection procedures of VGI are often ill-defined.
- In the broader sense, VGI can be defined to include data that we "volunteer" to companies.
- we must be mindful of the academic, legal, political, and environmental barriers for the development of open GI systems.

Geographic Representation

- Geographic representation is concerned with the Earth's surface or near-surface (≈critical zone) at scales from architectural to the global.

- The key GI representation issues are what to represent and how to represent it.

- Many plans for the real world can be tried out first on models or representation.

Geographic Representation

- The two fundamental ways of representing geography are discrete objects (離散物件) and continuous fields (連續場).

Discrete Objects

- The discrete object view represents the geographic world as objects with well-defined boundaries in otherwise empty space.

- One characteristic of the discrete object view is that objects can be counted.

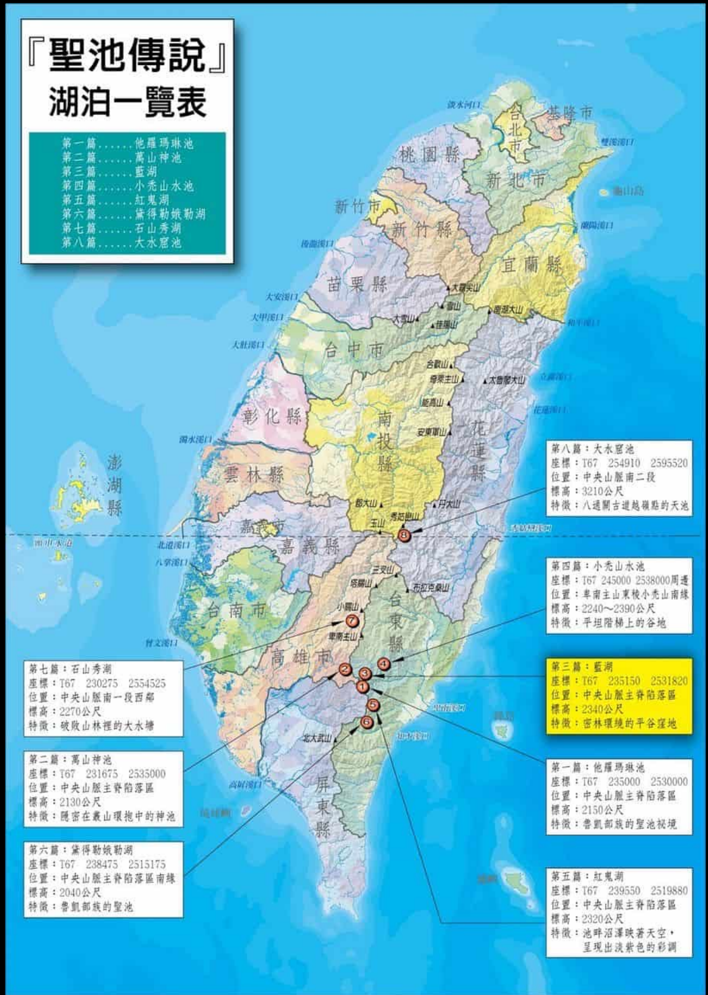
Continuous Fields

- In this view the geographic world can be described by a number of variables, each measurable at any point on the Earth's surface and changing in value across the surface.

- The continuous field view represents the real world as a finite number of variables, each one defined at every possible position.

- The variable may be nominal (類別), ordinal (序位), interval (間隔), ratio (比例), or cyclic (循環). A vector field (向量場) assigns two variables, magnitude and direction. Fields of only one variable are termed scalar field (純量場).

Continuous Fields



https://sunriver.com.tw/

Table 3.2 A scale of lakeness suitable for defining lakes as a continuous field.

Lakeness	Definition
1	Location is always dry under all circumstances.
2	Location is sometimes flooded in spring.
3	Location supports marshy vegetation.
4	Water is always present to a depth of less than 1 m.
5	Water is always present to a depth of more than 1 m.

Instead of counting, our strategy would be to lay a grid over the map and assign each grid cell a score on the lakeness scale. The size of the grid cell would determine how accurately the result approximated the value we could theoretically obtain by visiting every one of the infinite number of points in the state.

W1

Advantages of Learning GIS

Mapping without GIS

- Time-consuming
- Data is not reusable
- Lack of flexibility to change the content

Use GIS for mapping

- Create a map instantly
- Data is reusable
- High flexibility to change the content

Digital Presentation

- Digital data are easy to copy, they can be transmitted at close to the speed of light, they can be stored at high density in very small space, and they are less subject to the physical deterioration that affects paper and other physical media.

- Data in digital form are easy to transform, process, and analyze.

- Computers represent phenomena as binary digits. Every item of useful information about the Earth's surface is ultimately reduced by a GI database to some combination of 0s and 1s.

- Data in digital form are all discrete.

Digital Presentation

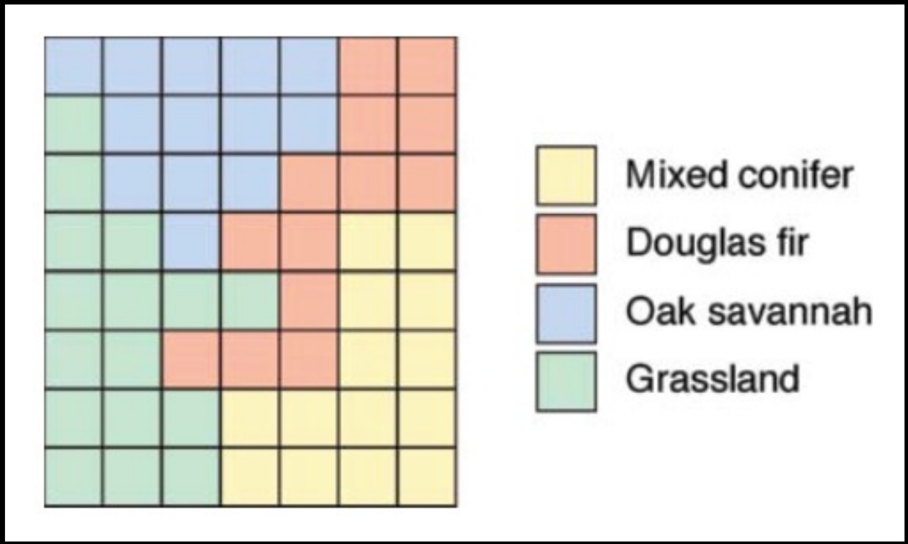
- Raster and vector are two methods of representing geographic data in digital computers.

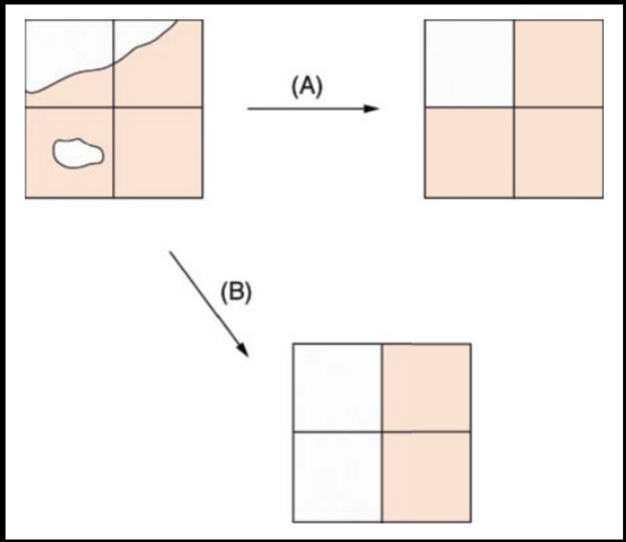
- In principle, both can be used to code both fields and discrete objects, but in practice a strong association exists between raster and fields, and between vector and discrete objects.

- “Raster is vaster, but vector is correcter.”

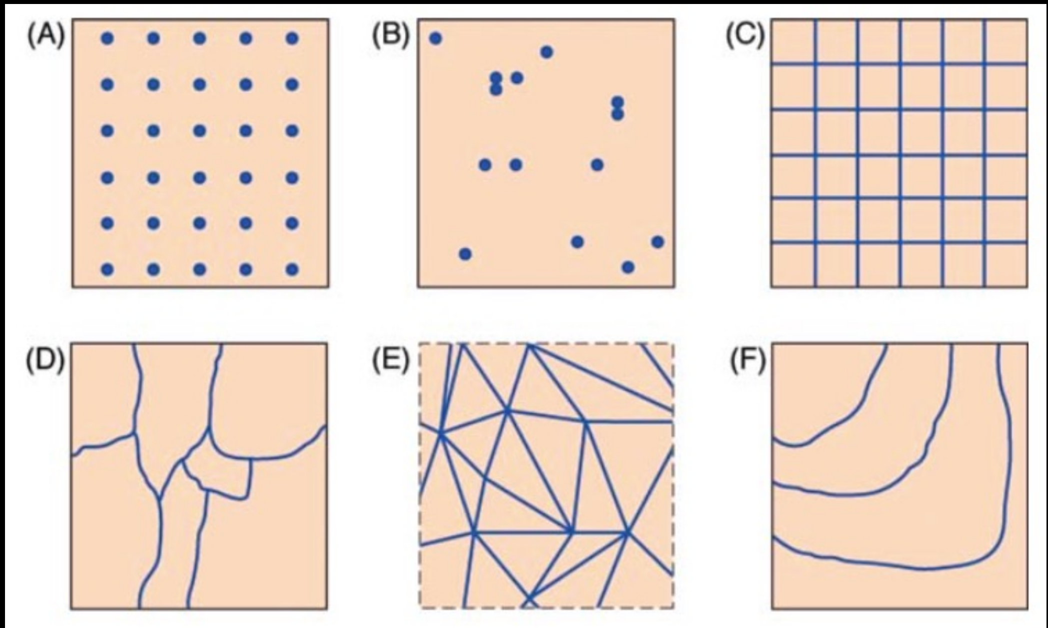
Raster Data

- In a raster representation, space is divided into an array of rectangular cells (sometimes called pixels) and assign attributes to the cells.
- One of the most common forms of raster data is remote-sensing images.
- Because of the distortions by flattening, the cells never be perfectly equal in shape or area on the Earth’s surface.





Representing Continuous Fields



(A) Capturing the value of the variable at each of a grid of regularly spaced sample points

(B) Capturing the value of the fi eld variable at each of a set of irregularly spaced sample points.

(C) Capturing a single value of the variable for a regularly shaped cell.

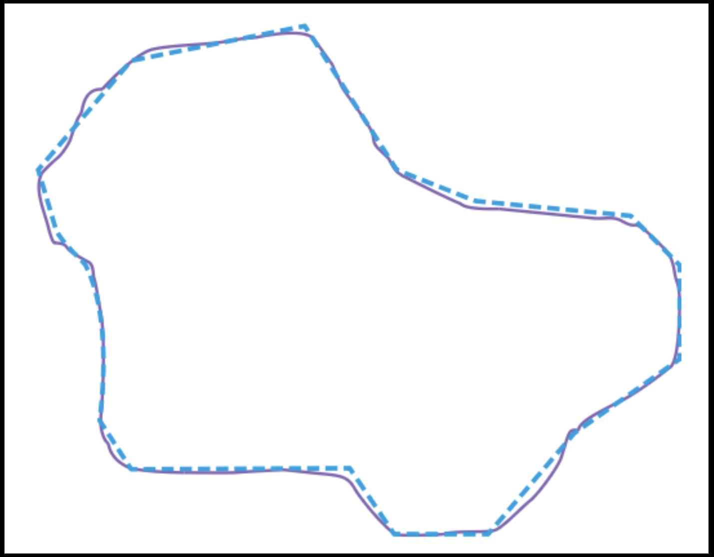
(D) Capturing a single value of the variable over an irregularly shaped area.

(E) Capturing the linear variation of the field variable over an irregularly shaped triangle.

(F) Capturing the isolines of a surface as digitized lines.

Vector Data

- In a vector representation, all lines are captured as points connected by precisely straight lines.
- An area is captured as a series of points or vertices (頂點) connected by straight lines to form a polygon (多邊形 / 面狀要素).
- Polyline (折線) is coined to describe a curved line represented by a series of straight segments connecting vertices.
- We can measure the area and perimeter.



Maps and Cartography

- A map is the graphic representation of the geographical setting.
- Cartography is the making and study of maps in all their aspects.
- The most fundamental function of maps is to bring things into view.
- All maps have one thing in common: to add the geographic understanding of the viewer.
- All geographical maps are reductions. Thus, they are smaller than the region they portrays.
- All maps are abstractions of reality.

W1

Maps and Cartography

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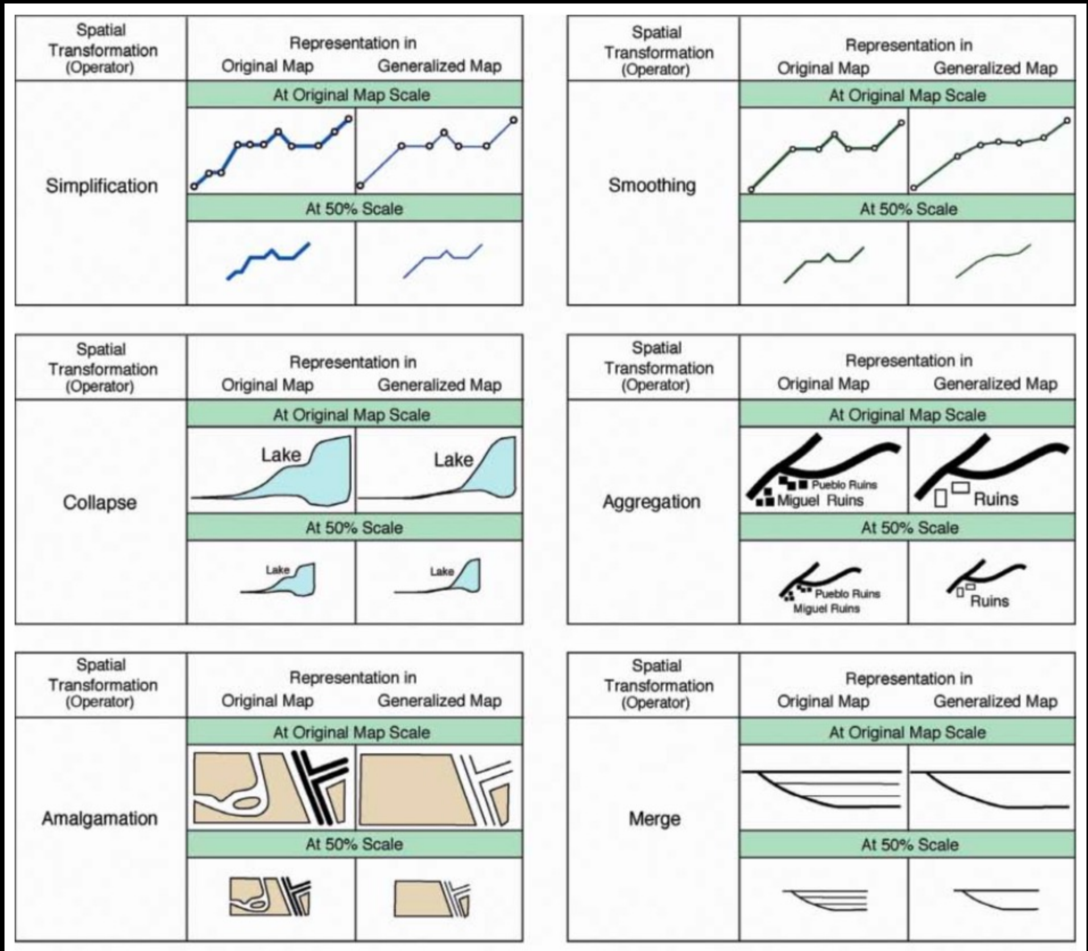
Generalization (概括化)

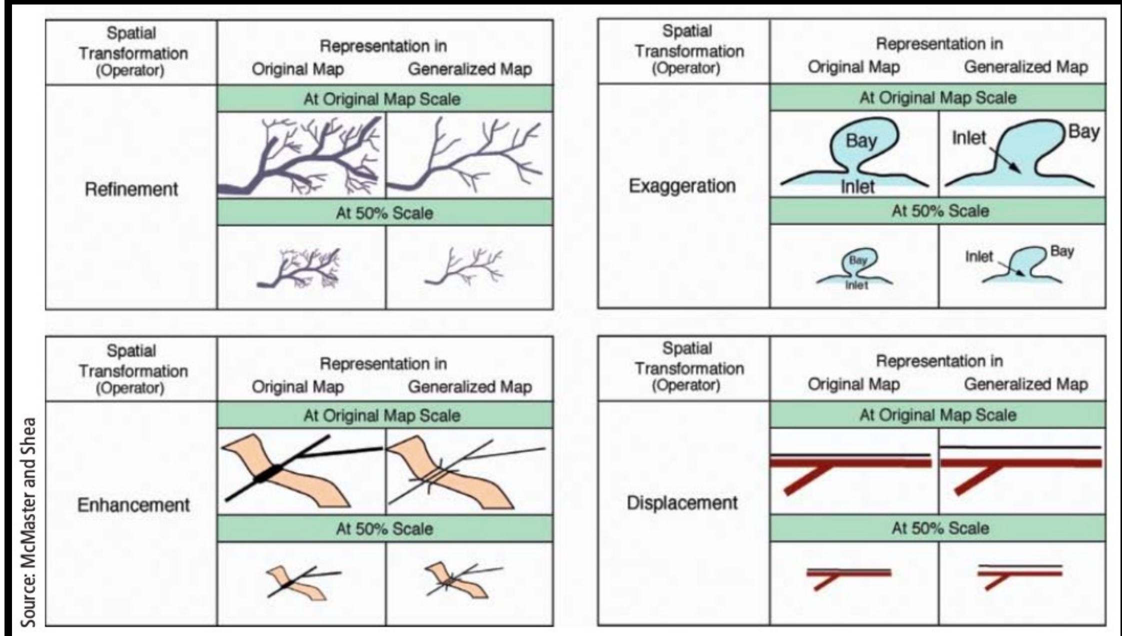
- All maps are abstractions of reality.
- A map’s specification defines how real features on the ground are selected for inclusion on the map.
- A geographic database cannot contain a perfect description; instead, its contents must be carefully selected to fit within the limited capacity of computer storage devices.

McMaster and Sea's Simplification



- Simplification (簡化), for example, by weeding out points in the outline of a polygon to create a simpler shape
- Smoothing (平滑), or the replacement of sharp and complex forms by smoother ones
- Collapse (降維), or the replacement of an area object by a combination of point and line objects
- Aggregation (聚合), or the replacement of a large number of distinct symbolized objects by a smaller number of new symbols
- Amalgamation (面合併), or the replacement of several area objects by a single area object
- Merging (線合併), or the replacement of several line objects by a smaller number of line objects

McMaster and Sea's Simplification



- Refinement (細化), or the replacement of a complex pattern of objects by a selection that preserves the pattern's general form
- Exaggeration (誇張), or the relative enlargement of an object to preserve its characteristics when these would be lost if the object were shown to scale
- Enhancement (強化), through the alteration of the physical sizes and shapes of symbols
- Displacement (位移), or the moving of objects from their true positions to preserve their visibility and distinctiveness

Douglas-Poiker Algorithm

- One of the most common forms of generalization of GI is the process known as weeding (點位刪減), which is invented by David Douglas and Tom Poiker.

- Weeding is the process of simplifying a line or an area by reducing the number of points in its representation.

- Remove points, which exceed the user-defined tolerance, in each iteration.